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Use Case 1 — Load and Clean Chicago Crime Data

Objective: Build a data ingestion pipeline to read, clean, and insert crime data into MySQL.

Tasks

1. Load the dataset (CSV) into a Pandas DataFrame

Inspect the first 10 rows; show schema and data types.

How many columns and rows are there?

```
First 10 rows:
      id case_number  ... longitude      location
0  10000000    SS442871  ... -87.673423  (41.71608165, -87.67342345)
1  10000001    WW348136  ... -87.626794  (41.94851207, -87.62679436)
2  10000002    GG525832  ... -87.629846  (41.96394084, -87.62984598)
3  10000003    NC685131  ... -87.607711  (41.77004756, -87.60771146)
4  10000004    YT673638  ... -87.602075  (41.72772815, -87.60207475)
5  10000005    CR348066  ... -87.582994  (41.85177018, -87.58299423)
6  10000006    XE039726  ... -87.522090  (41.98069888, -87.52208991)
7  10000007    QK559272  ... -87.621489  (41.74574749, -87.62148947)
8  10000008    YD115267  ... -87.709245  (41.85296539, -87.70924486)
9  10000009    DI878822  ... -87.516485  (41.72356216, -87.51648472)

[10 rows x 22 columns]
```

2. Clean the dataset

Convert the Date column into datetime format.

Identify and handle missing values for key fields.

Standardize categorical fields (strip whitespace, unify case).

```
# 2. Cleaning csv
# Convert Date column into datetime format
df['Date'] = pd.to_datetime(df['date'], errors='coerce')

# Identify and handle missing values for key fields
df['location_desc'] = df['location_desc'].fillna('Unknown')
df['block'] = df['block'].fillna('Unknown')
df['primary_type'] = df['primary_type'].fillna('Unknown')
df['description'] = df['description'].fillna('Unknown')

# Standardize categorical fields (strip whitespace, unify case)
df['primary_type'] = df['primary_type'].astype(str).str.strip().str.upper()
df['description'] = df['description'].astype(str).str.strip().str.upper()
df['location_desc'] = df['location_desc'].astype(str).str.strip().str.upper()
```

3. Generate new features

Extract Year, Month, and DayOfWeek from the Date field in chicago_crime_dataset.csv.

```
df['Year'] = df['Date'].dt.year
```

```
df['Month'] = df['Date'].dt.month
```

```
df['DayOfWeek'] = df['Date'].dt.day_name()
```

```
# 3. Generate new features
df['Year'] = df['Date'].dt.year
df['Month'] = df['Date'].dt.month
df['DayOfWeek'] = df['Date'].dt.day_name()
```

4. Use NumPy

Calculate the percentage of missing values per column.

Can you drop any columns with > 50% missing?

Missing values percentage per column:

```
{'id': 0.0, 'case_number': 0.0, 'date': 0.0, 'block': 0.0, 'iucr_code': 0.0, 'primary_type': 0.0, 'description': 0.0, 'location_desc': 0.0, 'arrest': 0.0, 'domestic': 0.0, 'beat_num': 0.0, 'district_code': 0.0, 'ward_no': 1.85, 'community_code': 1.8, 'fbi_code': 0.0, 'x_coordinate': 3.35, 'y_coordinate': 3.2, 'year': 0.0, 'date_of_update': 0.0, 'latitude': 3.65, 'longitude': 3.2, 'location': 3.7, 'Date': 0.0, 'Year': 0.0, 'Month': 0.0, 'DayOfWeek': 0.0}
```

Columns with > 50% missing values: []

5. Insert cleaned data into MySQL

Design a crime table schema in MySQL.

Use SQLAlchemy or PyMySQL to write data.

```
# 5. Insert cleaned data into SQLite
os.makedirs(DB_DIR, exist_ok=True)
conn = sqlite3.connect(DB_PATH)

# Save main crime dataset into SQLite
df_to_save = df.copy()
if 'date' in df_to_save.columns and 'Date' in df_to_save.columns:
    df_to_save = df_to_save.drop(columns=['date'])
if 'year' in df_to_save.columns and 'Year' in df_to_save.columns:
    df_to_save = df_to_save.drop(columns=['year'])
df_to_save['Date'] = df_to_save['Date'].astype(str)
df_to_save.to_sql("crimes", conn, if_exists="replace", index=False)
```

Questions

How many unique crime types exist in the dataset?

Are there anomalies in date formats or obvious outliers?

How many unique crime types exist in the dataset? 16

Are there anomalies in date formats? 0 invalid dates found.

Graphs, Tables And Charts(If any)

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Dataset Summary

Total Rows: 2000

Total Columns: 22

Unique Crime Types: 16

Invalid Date Formats: 0

Missing Data Analysis

Column Name	Data Type	Missing Percentage (%)	More than 50% Missing?
id	int64	0.0%	No
case_number	object	0.0%	No
date	object	0.0%	No
block	object	0.0%	No
iucr_code	int64	0.0%	No
primary_type	object	0.0%	No
description	object	0.0%	No
location_desc	object	0.0%	No
arrest	bool	0.0%	No
domestic	bool	0.0%	No
beat_num	int64	0.0%	No
district_code	int64	0.0%	No
ward_no	float64	1.85%	No
community_code	float64	1.8%	No
fbi_code	object	0.0%	No
x_coordinate	float64	3.35%	No
y_coordinate	float64	3.2%	No
year	int64	0.0%	No
date_of_update	object	0.0%	No
latitude	float64	3.65%	No
longitude	float64	3.2%	No
location	object	3.7%	No
Date	datetime64[ns]	0.0%	No

First 5 Rows Preview

ID	Case Number	Date	Primary Type	Description	Arrest	Year	Month	DayOfWeek
10000000	SS442871	03/04/2022 07:47	THEFT	THEFT OVER \$500	False	2022	3	Friday
10000001	WW348136	06/27/2015 19:34	PUBLIC PEACE VIOLATION	DISORDERLY CONDUCT	False	2015	6	Saturday
10000002	GG525832	03/31/2021 10:20	VANDALISM	CRIMINAL DAMAGE	False	2021	3	Wednesday
10000003	NC685131	04/26/2019 13:29	MOTOR VEHICLE THEFT	AUTOMOBILE	False	2019	4	Friday
10000004	YT673638	09/07/2017 05:56	ASSAULT	SIMPLE	False	2017	9	Thursday